MILITARY SPECIFICATION

RELAY, ARMATURE, TYPE A-1

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of armature relay, designated type A-1.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-5015	Connectors; Electrical
MIL-D-5028	Drawings and Data Lists; Preparation of (for Engines, Accessories, and Other Auxiliary Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-M-7911	Marking, Identification of Aeronautical Equipment Assemblies, and Parts
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-S-7742	Screw-Threads, Standard, Aeronautical

DRAWINGS

Air Force-Navy Aeronautical Standard Drawing

AN3102	Connector, Receptacles - Box Mounting

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components.- The relay shall consist of a mounting base, a housing, electrical connectors, and an operating mechanism.

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3.2 Materials.- Materials shall be as specified herein. When materials are used that are not specifically designated, they shall be entirely suitable for the purpose intended.

3.2.1 Metals.- Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life.

3.2.2 Protective treatment.- When materials are used in the construction of the relay that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.3 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with AN. Bulletin No. 143, except as provided in the following paragraph.

3.2.3.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction.- The relay shall be designed to automatically disconnect an operating inverter in the event of voltage failure and connect a standby inverter into the system. The relay shall function as specified herein.

3.3.1 The relay shall perform the following specific operations:

(a) Transfer the direct-current power from the operating inverter to the stand-by inverter.

(b) Disconnect the alternating-current circuits of the operating inverter and connect the alternating-current circuits of the stand-by inverter.

(c) Disconnect the inverter load from the inverter in the event of failure of the operating inverter and also during switching operations.

(d) Energize an inverter failure warning light circuit at any time that proper alternating-current voltages are not being delivered by the operating inverter.

(e) Effect a change-over from the operating inverter to the stand-by inverter in the event the output voltage is less than the specified minimum operating voltage.

(f) Effect a change-over from the operating inverter to the stand-by inverter in the event the output exceeds the maximum operating voltage.

(g) Maintain a specified time delay period between the disconnection of a faulty inverter and the connection of an operating inverter into the system.

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(h) Maintain a starting time delay, effective only upon initial application of direct-current power to the inverter system, of sufficient time to allow electronic tubes to reach operating temperatures before a change-over is effected.

(i) Include means to permit the inverter change-over relay to switch from either inverter to the other in the event of overvoltage or undervoltage output of the operating inverter.

(j) Include a test circuit in order to obtain a functional test of both inverters.

3.3.2 Sequence of operation.-

3.3.2.1 Undervoltage.- A relay, hereinafter specified as alternating-current voltage relays, shall be connected across each alternating phase to ground in such manner that the relay coil will be energized by the out-put voltage of the operating inverter. These relays shall pick up when the applied voltage is 102.5 ±2.5V rms. The operation of these relays shall cause the inverter to be connected to the load and the inverter failure warning light to be disconnected. These relays shall drop out at a voltage of 92.5 ±2.5V rms and shall energize the inverter failure warning-light circuit and a thermal time-delay circuit. At the end of the specified time delay, the switch shall energize the direct-current power transfer relay, the alternating-current inverter transfer relay, and the time delay lock-in relays. The direct-current power transfer relay shall switch the direct-current power from the operating inverter to the stand-by inverter, the alternating-current inverter transfer relay shall disconnect the alternating-current circuits of the operating inverter and connect to the alternating-current circuits of the stand-by inverter, and the lock-in relay shall short circuit the thermal-switch contacts and open the time-delay heater circuits. Normal voltage, as specified herein, from the stand-by inverter shall cause the alternating-current voltage relays to pick up and thereby energize the relays to connect the alternating-current load circuit to the inverter and shall disconnect the failure warning light. A loss of proper output voltage of No. 2 inverter shall allow the alternating-current voltage relays to drop out thereby causing a similar sequence of operations to disconnect No. 2 inverter and reconnect No. 1 inverter to the system.

3.3.2.2 Overvoltage.- An alternating-current voltage relay shall be connected as specified for undervoltage and adjusted to pick up at 125 ±2.5V rms. Pick-up operation of this relay shall cause a sequence of operations to disconnect the operating inverter and connect the stand-by inverter into the electrical system. The overvoltage relay shall be connected to phase "C" of the inverter output. This sequence of operations shall utilize the same inverter-circuit control relays as are used for undervoltage change-over. Failure of both inverters due to either overvoltage or undervoltage shall cause the relay to cycle from one inverter to the other until the inverter system is disconnected from the aircraft power supply.

3.3.2.3 Direct-current power transfer relay circuit.- Separate direct-current power circuits shall be provided for the operating and stand-by inverters. The relay shall be suitably arranged to completely disconnect the operating inverter from the direct-current bus and connect the stand-by inverter to the direct-current bus. The relay contacts shall be rated at 25-amp motor load at 28.5V dc. This relay shall be arranged to start the No. 1 inverter upon application of direct-current power to the inverter system.

3.3.2.4 Alternating-current inverter transfer relay.- The alternating-current inverter transfer relay shall be connected to the alternating current-input terminals of the relay assembly and shall be arranged to switch 2 of the 3-phase circuits from the operating to the stand-by inverter. The relay contacts shall be rated at not less than 7.4 amp at 115V, 400 cycles, ac, resistive load.

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3.3.2.5 Alternating-current load relay.- The alternating-current load relay shall be connected to the alternating-current output terminals of the relay assembly and arranged in such a manner that the load will be connected only when the alternating-current input voltages are within the normal range as specified herein. The contacts shall be rated as specified in paragraph 3.3.2.4.

3.3.2.6 Initial time delay.- A suitable time-delay circuit shall be arranged to provide a time delay of 70 ±5 seconds at 24V dc upon the initial application of direct-current power to the inverter system. This delay circuit shall be arranged to lock-in at the expiration of the delay period, to de-energize the time-delay heater circuits and to connect the normal 4-second time delay circuits to the control relays.

3.3.2.7 Normal time delay circuits.- Operation of either the alternating-current overvoltage or undervoltage relay due to high or low voltages shall energize a suitable time-delay circuit arranged to provide a 4 ±1 second delay between the time of the operation of the alternating-current voltage relay and the completion of the change-over from the operating inverter to the stand-by inverter. Both the initial time-delay switc and the normal time-delay switch shall be calibrated at 24V dc and shall be compensated for temperature. Suitable provisions shall be made to permit the use of the normal time delay switch without the 70-second initial delay. The delay circuit shall be arranged to provide not less than 2 seconds "ON" time when automatically cycling from one inverter to the other.

3.3.3 Relays.-

3.3.3.1 Alternating-current undervoltage.- The alternating-current undervoltage relays may be either alternating-current coil or direct-current coil rectifier type. The relay shall be designed to pick up at 102.5 ±2.5V ac and shall drop out at 92.5 ±2.5V ac. Suitable contact pile-ups as required for the control of auxiliary relays shall be provided. Drop-out of either of these relays for a period in excess of the normal time delay specified herein shall cause change-over.

3.3.4 Test switch circuit.- The internal wiring to the alternating-current circuit connected to the "A" phase of the inverter output shall be arranged to provide an external switch circuit to remove the voltage from the "A" phase alternating-current undervoltage relay. Operation of this switch shall allow the relay to drop out thereby causing a change-over from the operating inverter to the stand-by inverter. Holding the test switch in the operating position shall cause the relay assembly to cycle from one inverter to the other.

3.3.5 Inverter failure warning light.- A circuit shall be provided within the inverter changeover relay assembly to energize a warning-light circuit at any time that either of the undervoltage relays are in the drop-out position or either of the over-voltage relays are in the pick-up position. The warning-light circuit shall remain energized until proper voltage is applied to the alternating-current voltage relays. This circuit shall permit a lamp load of not less than 2.5 amp, 28V dc to be energized at any time that the alternating-current voltage is outside the limits specified herein for normal voltage.

3.3.6 Normal voltage.- The normal alternating-current output voltage for inverters shall be 110 to 120V.

3.3.7 Normal direct-current voltage.- The normal direct-current input voltage shall be from 22 to 30V. All direct-current relays inclosed within the relay assembly shall pick up at a direct-current voltage not higher than 17V and shall drop out at a direct-current voltage not higher than 8V.

3.3.8 External connections.- Connection of the internal circuits of the relay assembly to the external wiring within the aircraft shall be made through not more than two box-type receptacles conforming to Drawing AN3102 and Specification MIL-C-5015.

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3.3.9 Mounting.- The mounting arrangements of the relay assembly shall be as specified in figure 1. The relay assembly shall be designed to operate satisfactorily when mounted in any position.

3.3.10 Container.- The entire relay assembly shall be enclosed in a leakproof corrosion-resistant metal container and pressurized with an inert gas or blend of gases containing not less than 10 percent helium by volume to a pressure greater than sea level atmospheric. Leakage of the pressurizing gas shall not exceed 1 micron cubic foot per hour. The exterior of the sealed container shall be suitably protected from corrosion in a salt fog atmosphere by painting or plating as required.

3.4 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Screw threads.- Screw threads shall be in accordance with Specification MIL-S-7742. Fit of screw threads shall conform with NF-2A for external threads and NF-3B for internal threads.

3.6 Performance.- The relay shall be capable of satisfactory operation after being subjected to the following conditions:

(a) Ambient temperatures ranging from -55° to +71°C.
(b) Exposure to salt sea atmosphere.
(c) Vibration incident to service use.
(d) Acceleration and shock forces incident to service use.

3.7 Identification of product.-

3.7.1 Nameplate.- A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the relay. The information marked in the spaces provided on the nameplate shall be in accordance with Specification MIL-M-7911.

Relay, Armature
Type A-1
Specification MIL-R-6518A
Stock No.(USAF or Navy as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trade-mark
US Property

3.8 Workmanship.-

3.8.1 General.- The relay, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alinement of parts and tightness of assembly screws, and bolts, et cetera.

3.8.2 Cleaning.- The relay shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of relays shall be classified as follows:

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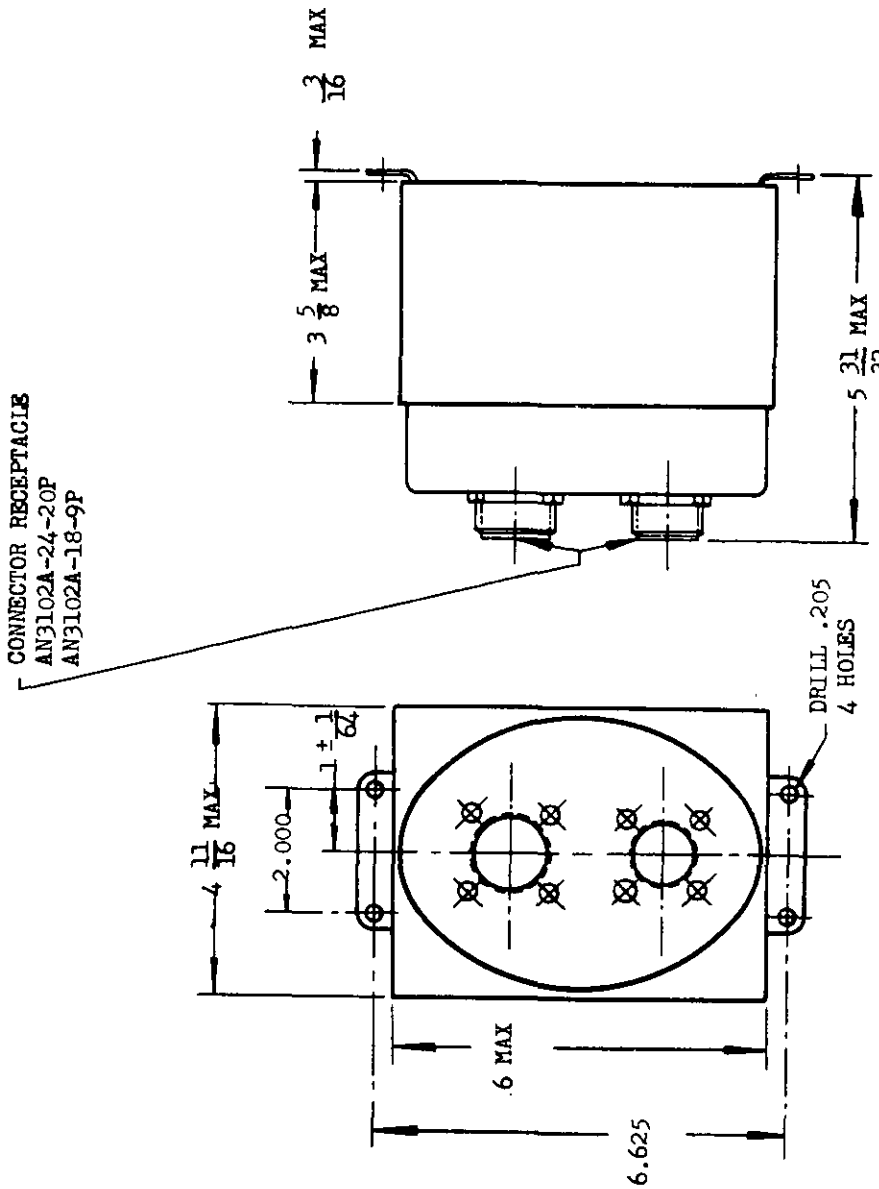


FIGURE 1. Mounting

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(a) Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those test accomplished on relays manufactured and submitted for acceptance under contract.

4.2 Qualification tests.-

4.2.1 Prior qualification.- Unless otherwise specified by the procuring activity, relays which have not previously passed a Qualification test, or which have passed the Qualification test and have been modified in any manner, shall satisfactorily pass a Qualification test prior to the acceptance of any relay.

4.2.2 Sampling instructions.- Qualification test samples shall consist of two relays. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from the qualifying activity. (See paragraph 6.3.)

4.2.3 Tests.- The Qualification tests shall consist of the tests specified under Inspection tests and the following tests.

4.2.3.1 Life cycle.- The relay assembly shall be connected in a suitable test set-up to provide the relay contact loadings specified herein. By means of overvoltage and undervoltage being applied to each alternating-current voltage sensing relay, in proper sequence, the relay assembly shall be caused to change-over from one inverter position to the other. One complete change-over from No. 1 position to No. 2 position then back to No. 1 position shall constitute one cycle. The relay shall be operated at room conditions and loads as specified for 10,000 complete cycles. Upon completion of this test the relay shall meet the requirements of the Insulation and Calibration tests.

4.2.3.2 High temperature.- The relay assembly shall be subjected to High temperature tests, Procedure I of Specification MIL-E-5272. The relay assembly shall be operated as specified for life cycle for 1 hour during each 24-hour period. At the conclusion of this test and while still at this temperature, the relay assembly shall meet the calibration test requirements.

4.2.3.3 Low temperature.- The relay shall be subjected to Low temperature tests, Procedure II of Specification MIL-E-5272. At the end of the test period, and while maintaining the low temperature, the relay shall meet the calibration test requirements and operated 1 hour as specified for Life cycle.

4.2.3.4 Salt spray.- The relay assembly, with appropriate plugs mated in the receptacles, shall be subjected to Salt spray tests, Procedure I of Specification MIL-E-5272 for 100 hours. At the end of this period, the relay assembly shall not show evidence of corrosion that would cause malfunction or failure.

4.2.3.5 Vibration.- The relay assembly shall be subjected to vibration frequencies from 6,000 to 15,000 cpm and amplitudes not to exceed 0.002 inch, total excursion. During this vibration range, there shall be no malfunctioning, inadvertent operation, or other signs of failure of the relay assembly. The relay shall then be subjected to vibration frequencies of 9,000 cpm and amplitude of 0.002 inch for 6 hours. Following this test, the relay shall be shock-mounted with appropriate sized vibration dampers. The shock-mounted relays shall be subjected to a vibration frequency varying 5 to 55 cps at an amplitude of 0.60 inch for a period of 6 hours. Following these tests, the relay assembly shall be operated for 100 cycles as specified for Life cycle test.

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4.2.3.6 Shock.- The relay assembly shall be subjected to Shock tests, Procedure I of Specification MIL-E-5272. The shock shall be equal to 25g. The relay shall not show evidence of malfunction or inadvertent change-over.

4.2.3.7 Acceleration.- The relay shall be subjected to Acceleration tests, Procedure II of Specification MIL-E-5272. An acceleration of 20g with the relay mounted in any plane shall not cause failure or inadvertent change-over of the relay assembly.

4.3 Inspection tests.- The Inspection tests shall consist of Individual tests.

4.3.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.3.2 Individual tests.- Each relay shall be subjected to the following tests.

4.3.2.1 Examination of product.- Each relay shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.3.2.2 Insulation tests.- The insulation of the electrical circuits within the container shall be determined by means of a variable direct-current voltage together with a series microammeter and suitable blocking resistors so arranged that the maximum current within the test circuit is 100 microamperes. The resistance of the insulation of the specified test voltage shall be determined from the dielectric current indicated on the microammeter. The dielectric current shall not exceed the value specified herein. Circuits connected to the alternating-current inverter input connections shall be subjected to a test voltage of 2,000V dc. Circuits connected to the 28V dc input connections shall be subjected to a test voltage of 1,000V ac. At the end of 60 seconds, the dielectric current shall not exceed 10 microamperes.

4.3.2.3 Operation.- Each relay shall be connected to a test set-up simulating the typical aircraft inverter installation and caused to operate not less than 100 cycles of change-over, from the operating inverter to the stand-by inverter and back to the operating inverter. Following this test, the relay assembly shall be tested for calibration as specified herein.

4.3.2.4 Calibration.- The relay shall be connected to circuits suitable for operation of the alternating-current voltage relays and the time-delay switches. Suitable circuits to sense the operation of the elements shall be arranged. The pick-up voltage of the alternating current overvoltage relay, the pick-up and drop-out voltage of the alternating current, undervoltage relay, the delay interval of both the initial delay and normal delay switches shall be obtained. These values shall be as specified herein.

4.3.2.5 Gas leakage.- The leakage of the pressurizing gas from the sealed container of each relay shall be determined by the Mass Spectrometer method of leak detection. The leakage shall not exceed the value specified herein.

4.3.3 Rejection and retest.- Relays which have been rejected may be reworked or replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

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4.4 All parts, specimens, or assemblies destroyed in making tests required by this specification and drawings, to determine compliance therewith, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 The relays shall be packaged and packed for shipment in accordance with group II of Specification MIL-P-6064 in a type I, class II container.

5.3 Marking of shipments.- Marking shall be in accordance with the requirements of Specification MIL-P-6064 and as specified herein.

5.3.1 Packages.- Each interior package shall be marked with the following information:

Relay, Armature, Type A-1
Specification MIL-R-6518A
Stock No. (USAF or Navy, as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Date of manufacture
Contract or Order No.
Quantity
Name of manufacturer
Name of contractor (if different from manufacturer)

IF IN STORAGE AFTER (DATE)*, THE RELAY SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

*This date shall be 3 years from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

5.3.2 Precautionary marking.- The following precautionary marking shall appear on two opposite sides of each interior package, whenever practicable, depending on the size of the carton and shall also appear on the exterior shipping container:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use.- The type A-1 armature relay covered by this specification is intended for use with 28V d-c input, 115V, 3-phase, 400-cycle, a-c output inverters to provide automatic change-over from one inverter to the second inverter in event of inverter failure thus affording a continuous supply of a-c power as long as either inverter remains operable, without requiring attention by operating personnel.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.3 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and

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manufacturers are urged to communicate with the Commanding General, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio; the activity responsible for qualification, with a copy to the other Service, and arrange to have the products that they propose to offer to the Army, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the above designated activity responsible for qualification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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